Pulseq on Philips II: Another Approach

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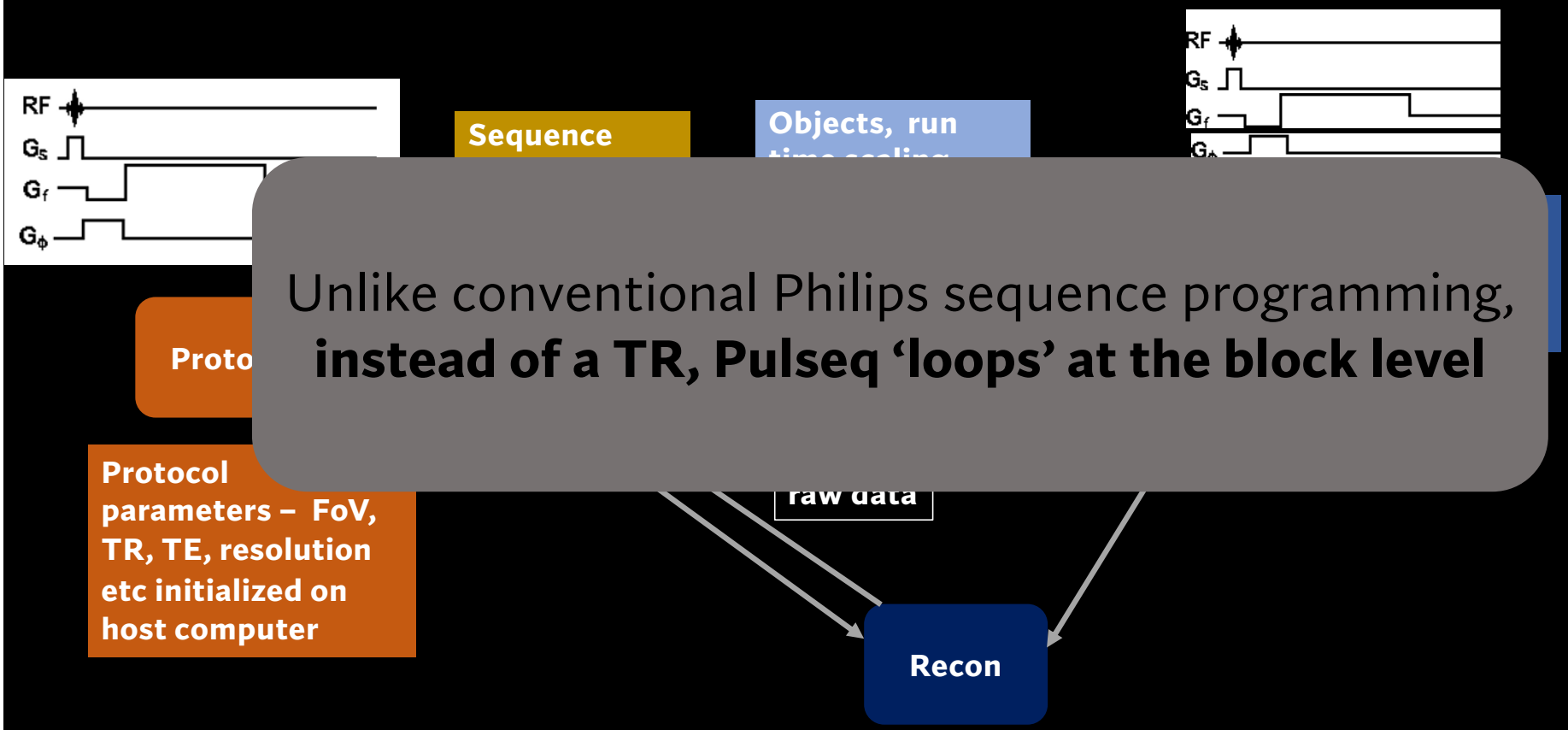
Motivation & Introduction

- Pulseq offers the potential to streamline the implementation of consistent pulse sequences across various scanners, yet it had not been integrated into the Philips platform until 2023.

- Implementing Pulseq sequences on Philips systems presents certain system constraints, while still satisfying system RF and gradient safety checks:
  - High RF and ADC deadtime/Ringdown time.
  - The requirement for arbitrary gradients to start and end at zero.
  - The capability to dynamically adjust the number of RF pulse shapes during execution.

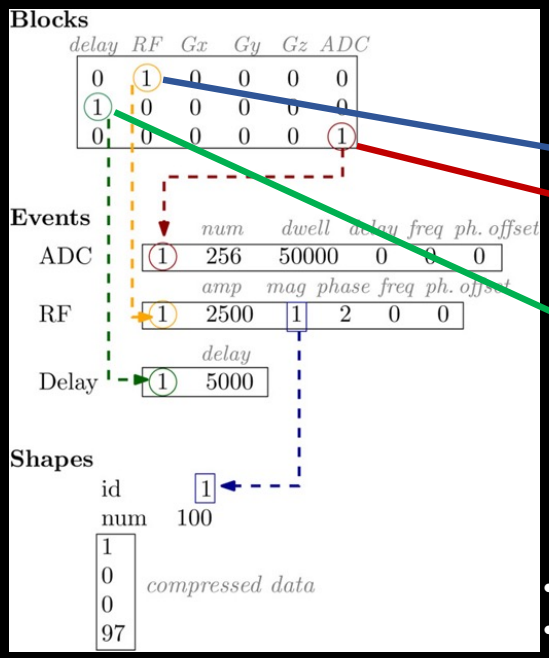
Philips sequence programming architecture

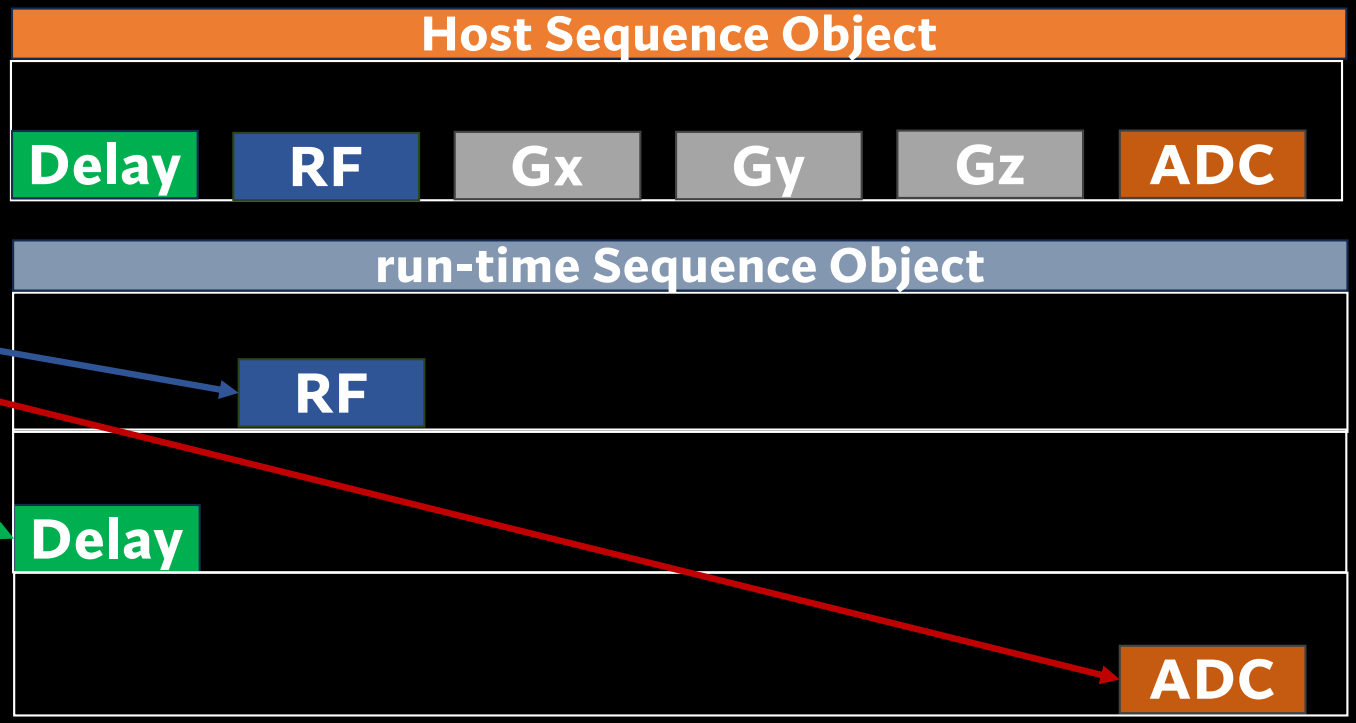


Unlike conventional Philips sequence programming, **instead of a TR, Pulseq 'loops' at the block level**

https://www.cis.rit.edu/htbooks/mri/inside.htm

Pulseq blocks mapping to Philips sequence objects

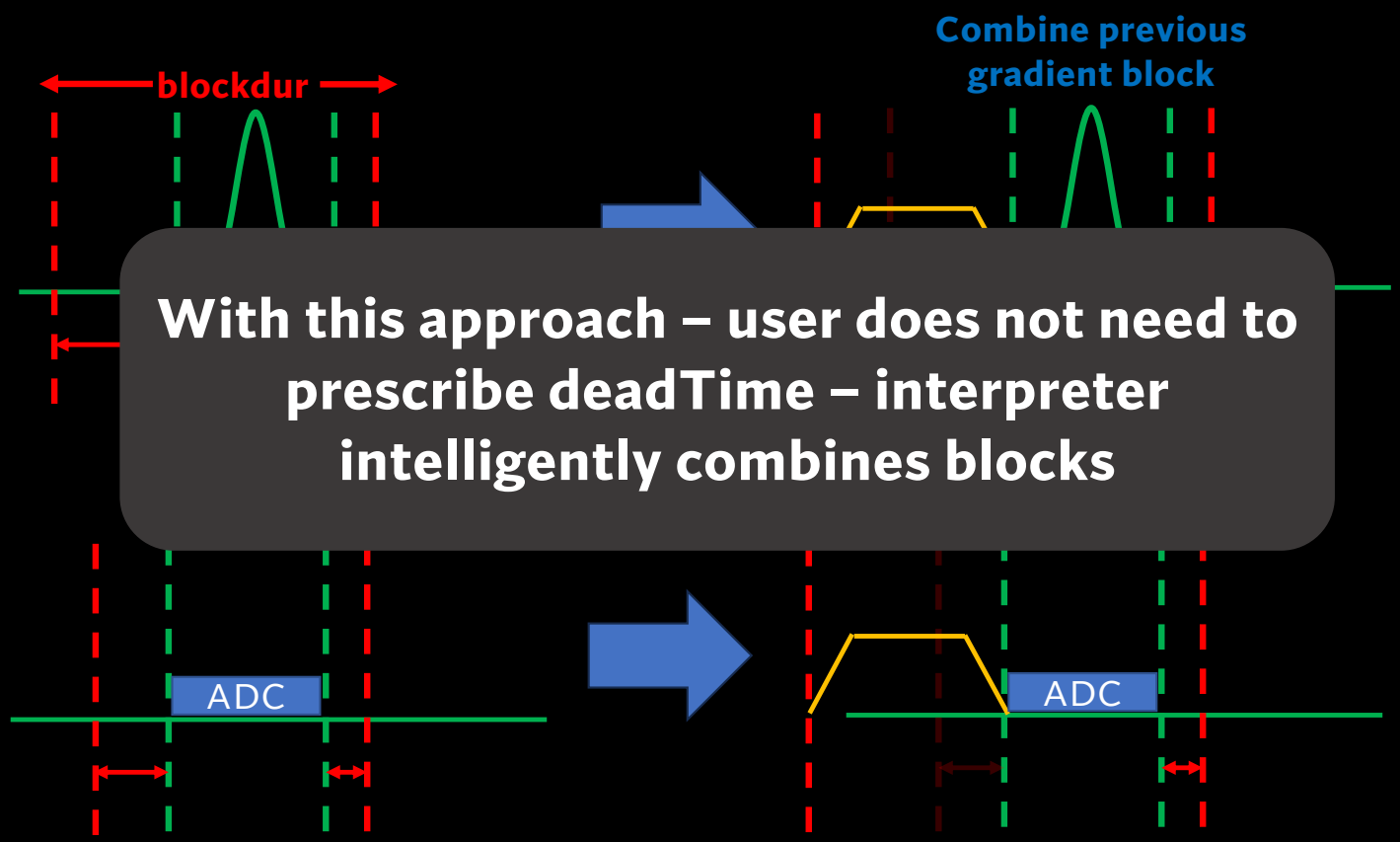




- **Update Object attributes during run time**
- **Not all attributes are run time** – interpreter smartly initializes some attributes on Host
- Gradient delay blocks can be executed without any timing constraint, but RF & ADC need to accommodate **deadTime/ringdownTime**

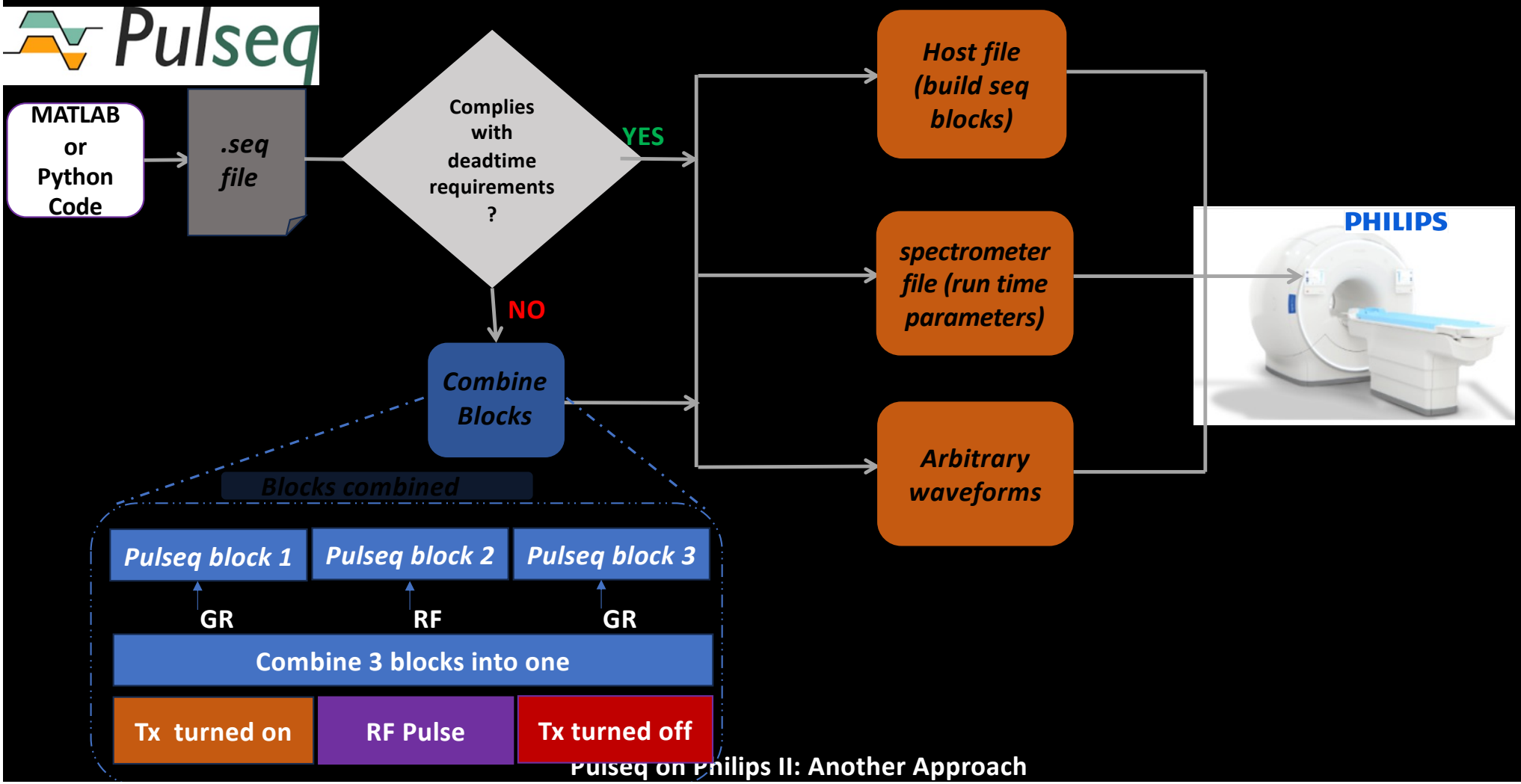
Magnetic Resonance in Med, Volume: 77, Issue: 4, Pages: 1544-1552, First published: 07 June 2016, DOI: (10.1002/mrm.26235)

Deadtime/Ringdown time for RF & ADC – combine blocks



With this approach – user does not need to prescribe deadTime – interpreter intelligently combines blocks

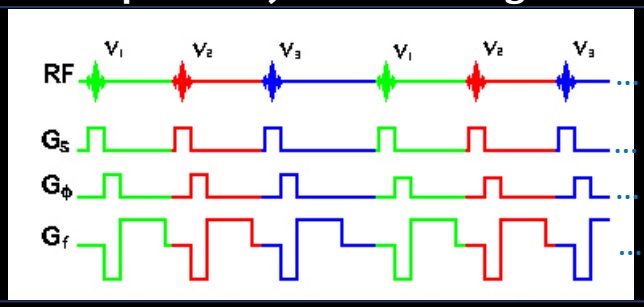
Pulseq to Philips (P2P) Block diagram



PNS, SAR & HW constraint safety checks

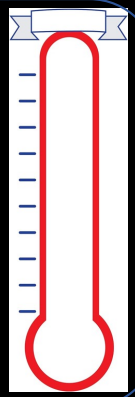
Output files from interpreter

One sequence object – combining 100 blocks



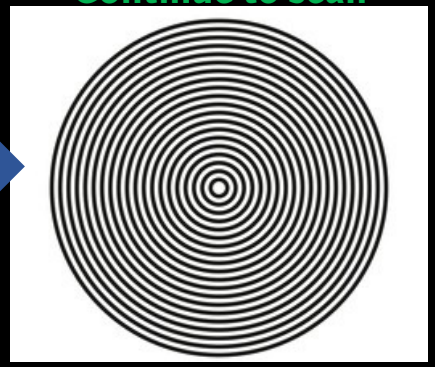
PNS and SAR eval for N iterations

- G_{max}
- Dwell time
- Filter freq
- Number of unique RF shapes



?

Continue to scan



Conflict raised

https://www.cis.rit.edu/htbooks/mri/chap-8/chap-8-h5.htm

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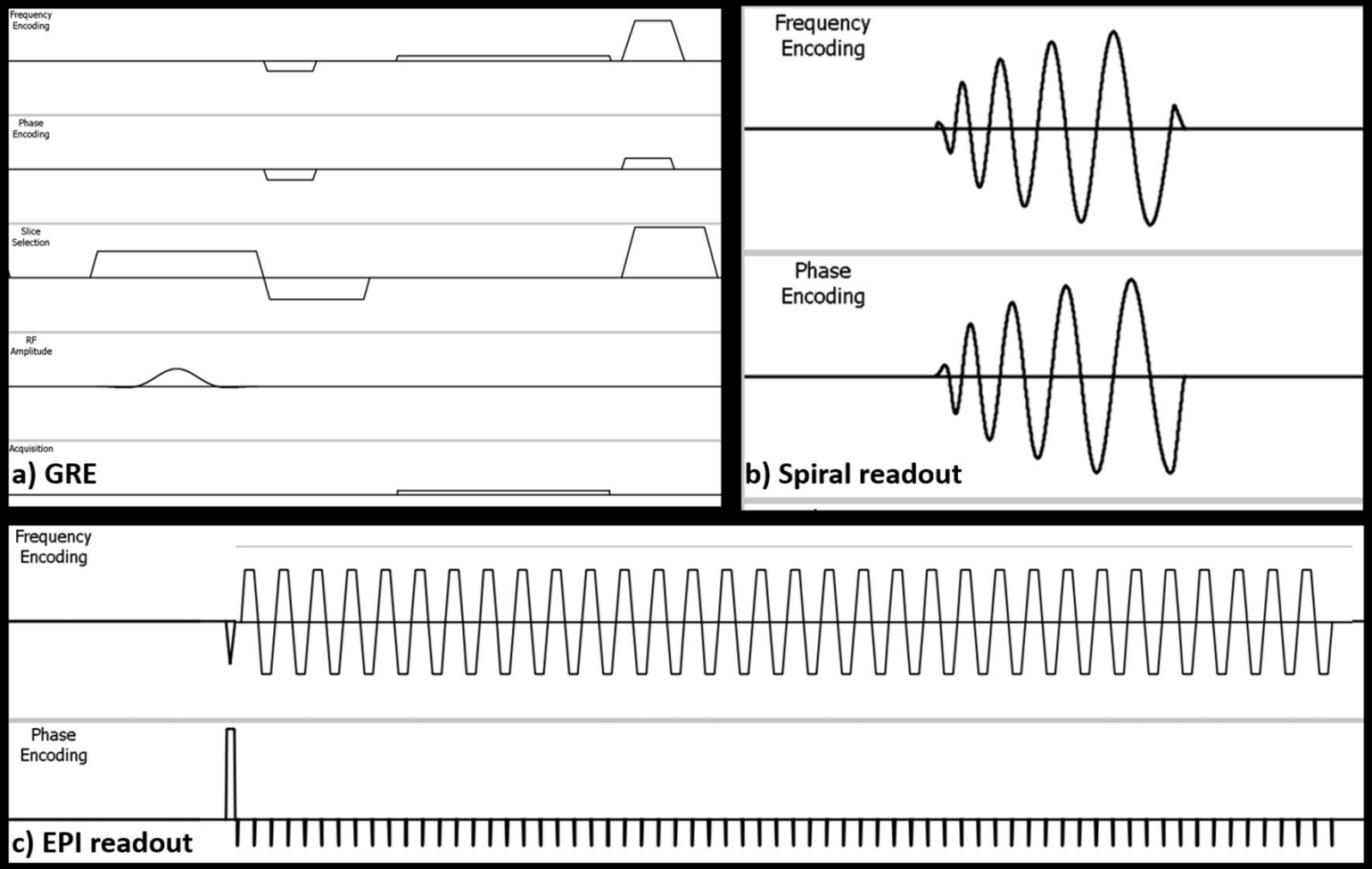
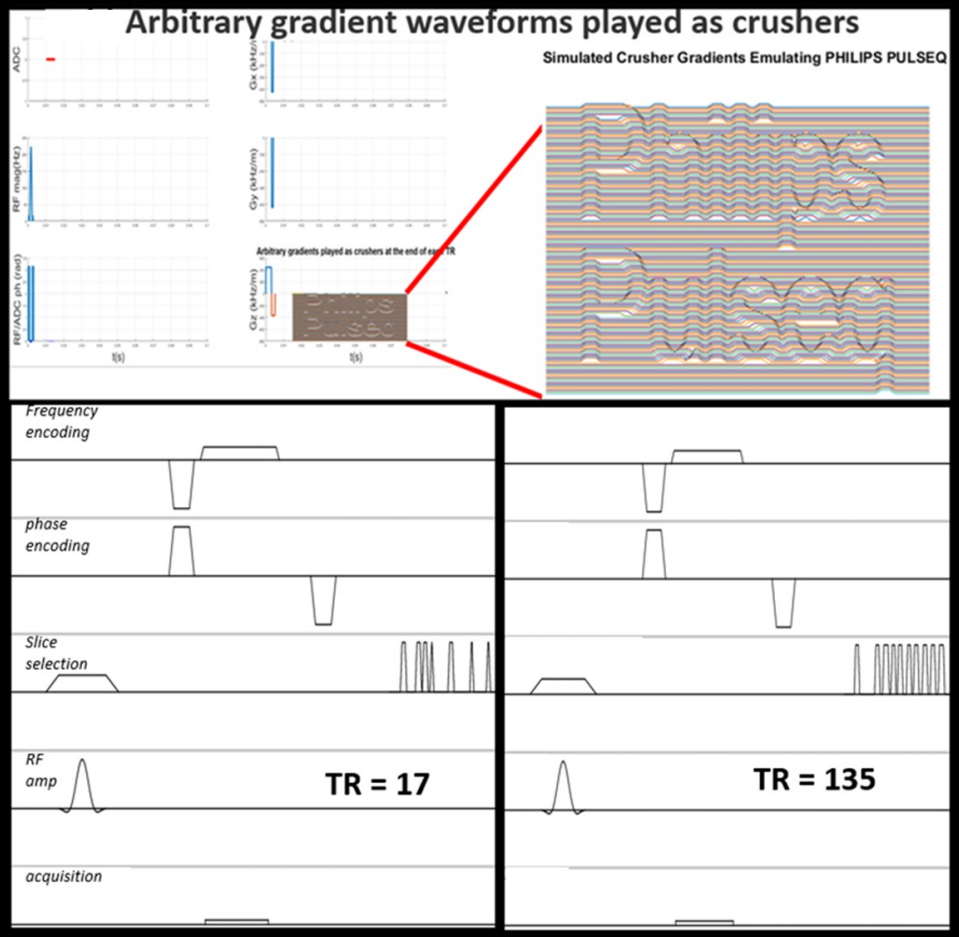
Different gradient waveforms captured from scanner graphical viewer

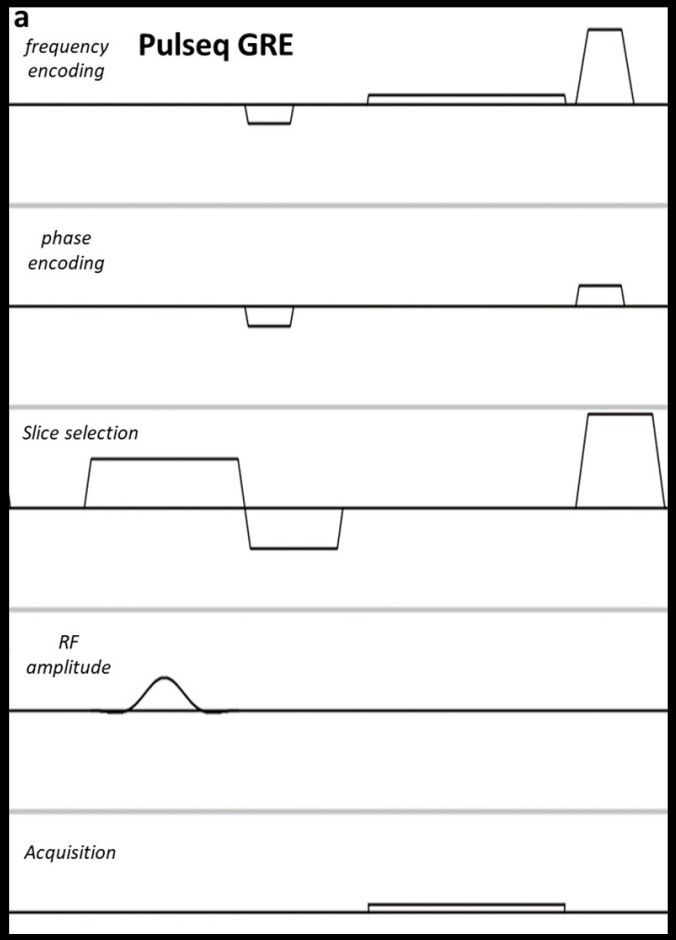


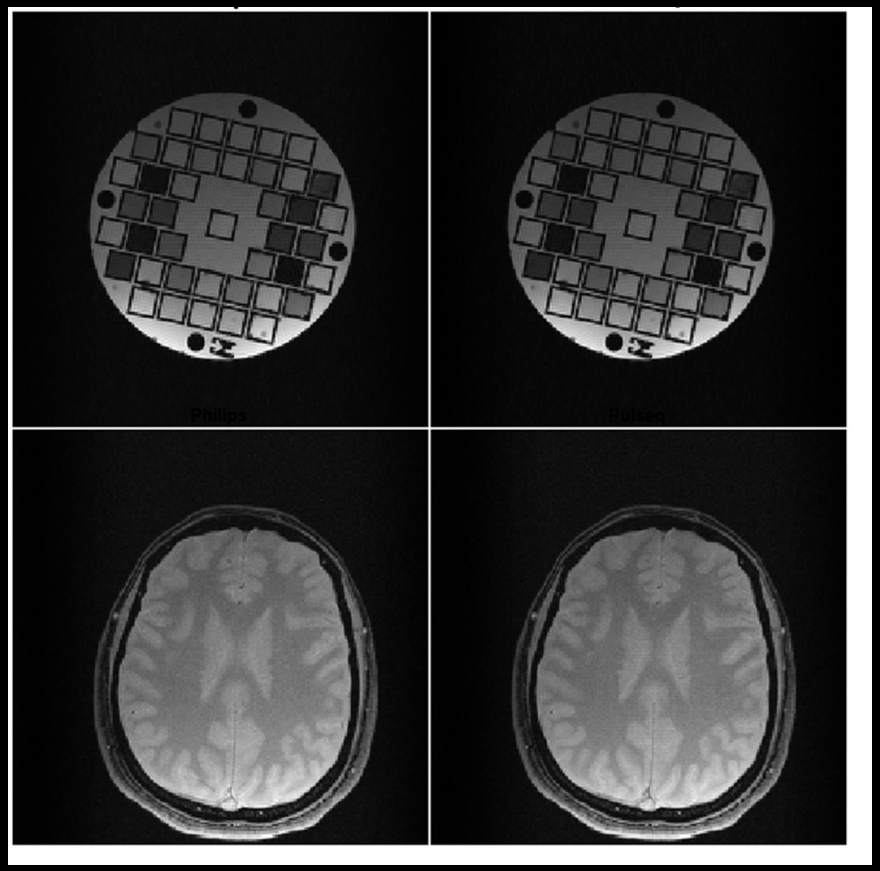
Arbitrary gradient waveforms

- **Any gradient and RF shape designed** for Philips system hardware constraints (Gmax, Slew rate, dwell time etc.) can be played
- Crusher gradients emulating “Philips Pulseq” simulated and played in a gradient echo sequence
- Two TRs – 17 & 135 from vendor provided graphical Viewer



Pulseq GRE comparison with Philips





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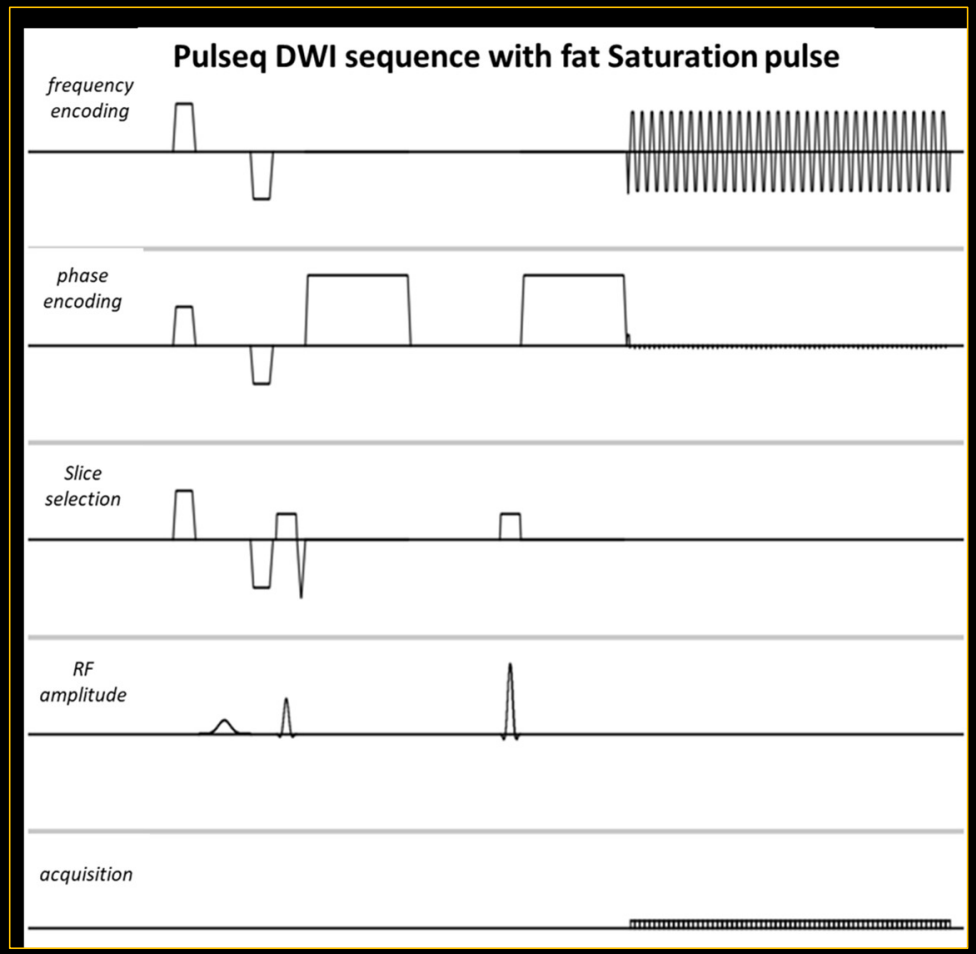
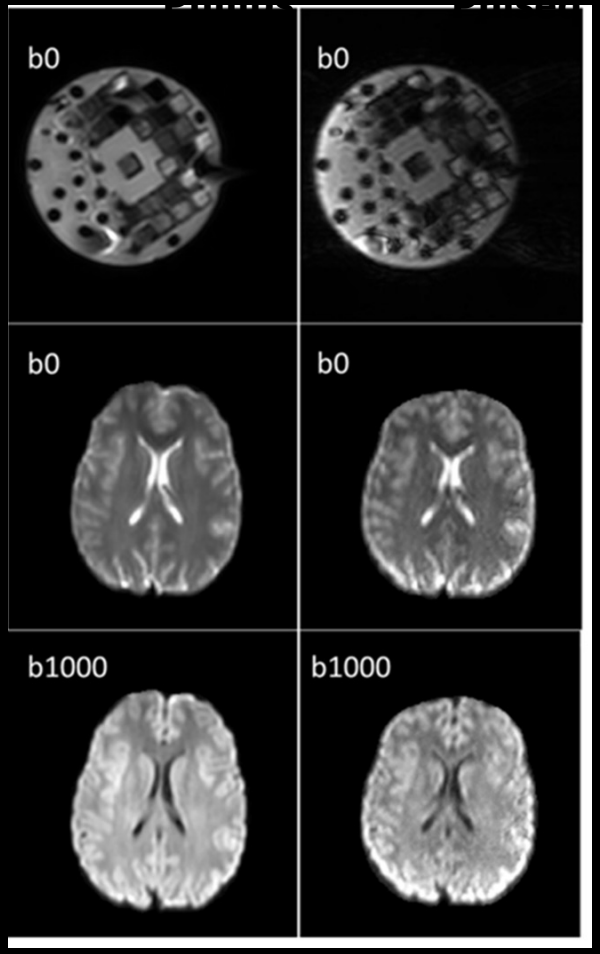
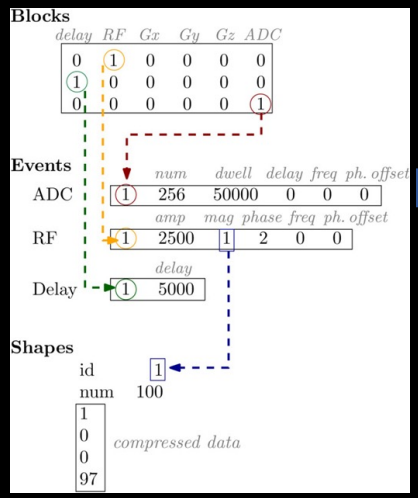
Pulseq DWI comparison with Philips





Arbitrary pulse sequence execution with pulseq on a Philips MRI scanner

P2P interpreter workflow from the user's perspective



seq2ppe.m

host.csv
runtime.csv
waveforms.csv

Patch

Raw data

Reconstructed Image

MATLAB Recon

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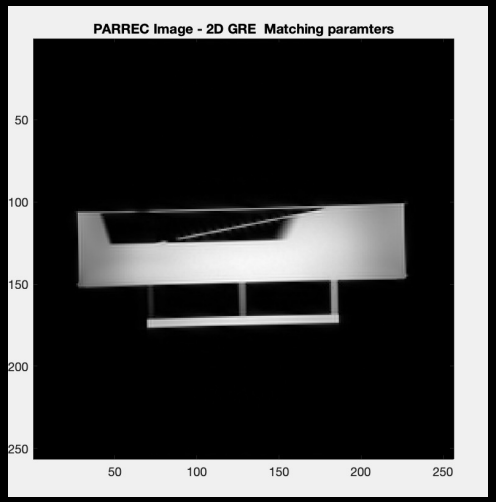
Discussion

- Any sequence can be played if the **deadTime and ringdownTime** are set according to Philips requirements
- Up to **256** distinct RF shapes
- Sequence with **100,000** blocks had **12%** CPU usage
- Current limitations and Future Work:
  - Blocks with consecutive RF cannot be combined
  - Geometric **scan planning is not supported** yet – acquisition happens w.r.t **iso-centre**
  - **RF block followed by ADC block** cannot be combined
  - For EPI sequence – Blocks incorporating readout, blip and ADC should iterate by simply altering the sign (readout time, blip duration and number of samples remain the same)

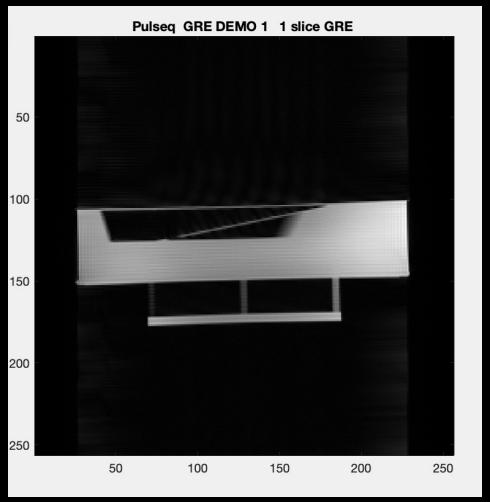
Where do things stand today?

- Shaik developed initial approach, ended work at end of 2024
- Saikat Sengupta (Vanderbilt) is picking up where Shaik left off, has re-built the patch and collected and reconstructed data
- Presently still in an early alpha state, working with 1-2 sites to enable specific driving projects
- Happy to share with Philips users with research agreements, with disclaimer

Philips GRE



Pulseq GRE Demo



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Acknowledgements

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Thank you!

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